

# Joonho Lee

Group Leader, Neuromeka AI Lab · Advisor, RIVR

On military duty until 2027.4.30. (Technical Research Personnel at Neuromeka)

Robotics researcher and systems integrator with field experience. PhD from the Robotic Systems Lab at ETH Zurich on legged robotics; now leading AI research for industrial robots at Neuromeka.

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## Experience

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|------------|---------------------------------------------------------|
| 2024 – now | <b>Neuromeka AI Group</b><br>Group Leader               |
| 2024 – now | <b>RIVR</b><br>Advisor, RL-based control and navigation |

## Education

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|-------------|---------------------------------------------------------------------------|
| 2018 – 2024 | <b>ETH Zurich, Robotic Systems Lab</b><br>PhD, Mechanical Engineering     |
| 2017 – 2018 | <b>ETH Zurich</b><br>MSc Robotics, Systems and Control · Rank 1/76        |
| 2012 – 2016 | <b>POSTECH</b><br>BSc Mechanical Engineering · Magna Cum Laude, Rank 2/17 |

## Funding

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|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 – 2029 | <b>AI-based medical and care humanoid robot: bidirectional personalised HRI, semantic navigation and bimanual autonomous manipulation</b><br>International Joint R&D Program, KIAT, Ministry of Trade, Industry and Energy<br>Planned the project and lead its execution (project lead, Neuromeka)<br>Partners: Georgia Tech (Sehoon Ha et al.), ETRI, KAIST<br>≈ US\$7M (KRW 10B) over 5 years |
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## Selected publications

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| 2026 | <b>Proprioceptive reflex learning for contact-rich manipulation</b><br>Quan Nguyen, Yunho Kim, Joonho Lee<br><a href="#">Submitted to ICRA 2027</a>                                                                    |
| 2026 | <b>Learning-augmented robotic automation for real-world manufacturing</b><br>Yunho Kim, Quan Nguyen, Taewhan Kim, Youngjin Heo, Joonho Lee<br><a href="#">arXiv:2604.22235 · under review at Nature Communications</a> |
| 2025 | <b>Learning fast, tool-aware collision avoidance for collaborative robots</b><br>Joonho Lee, Yunho Kim, Seokjoon Kim, Quan Nguyen, Youngjin Heo<br><a href="#">IEEE RA-L</a>                                           |
| 2025 | <b>Learning-based legged locomotion: State of the art and future perspectives</b><br>Sehoon Ha, Joonho Lee, Michiel van de Panne, Zhaoming Xie, Wenhao Yu, Majid Khadiv<br><a href="#">IJRR 44(8)</a>                  |
| 2024 | <b>Learning to walk in confined spaces using 3D representation</b><br>Takahiro Miki, Joonho Lee, Lorenz Wellhausen, Marco Hutter<br><a href="#">ICRA · Best Paper Award in Service Robotics</a>                        |

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| 2024 | <b>Scalable multi-robot cooperation for multi-goal tasks using reinforcement learning</b><br>Joonho Lee*, Tianxu An*, Marko Bjelonic, Flavio De Vincenti, Marco Hutter<br><a href="#">IEEE RA-L</a>                                         |
| 2024 | <b>Learning robust autonomous navigation and locomotion for wheeled-legged robots</b><br>Joonho Lee, Marko Bjelonic, Alexander Reske, Lorenz Wellhausen, Takahiro Miki, Marco Hutter<br><a href="#">Science Robotics 9(89)</a>              |
| 2024 | <b>When monkeys meet an ANYmal robot in the wild</b><br>Charlotte Canteloup, Joonho Lee, Samuel Zimmermann, Morgane Alvino, Markus Montenegro, Marco Hutter, Erica van de Waal<br><a href="#">bioRxiv 2024.08.13.607714</a>                 |
| 2022 | <b>Learning robust perceptive locomotion for quadrupedal robots in the wild</b><br>Takahiro Miki, Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, Marco Hutter<br><a href="#">Science Robotics 7(62)</a>                      |
| 2020 | <b>Learning quadrupedal locomotion over challenging terrain</b><br>Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, Marco Hutter<br><a href="#">Science Robotics 5(47)</a> · <a href="#">IEEE RAS TC-MBOR Best Paper Award</a> |

## Awards

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|------|------------------------------------------------------------------------------------------------------------|
| 2025 | <b>ETH Zurich Medal for outstanding doctoral thesis</b><br>ETH Zurich                                      |
| 2024 | <b>Best Paper Award in Service Robotics</b><br>IEEE ICRA                                                   |
| 2020 | <b>Best Paper Award, TC on Model-Based Optimization for Robotics</b><br>IEEE Robotics & Automation Society |
| 2019 | <b>ETH Zurich Medal for outstanding master's thesis</b><br>ETH Zurich                                      |
| 2019 | <b>Willi Studer Prize (Best Graduate), Robotics, Systems and Control</b><br>D-MAVT, ETH Zurich             |

## Invited talks & lectures

22 TALKS

|            |                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------|
| 2026.09.22 | <b>From Paper to Factory Floor: Building Smart Robots for Industry</b> – ME Seminar, KAIST (KR)   |
| 2025.11.28 | <b>Teaching Robots to Walk and Work</b> – CSE/GSAI Seminar, POSTECH (KR)                          |
| 2025.06.04 | <b>Training Robots to Move and Work</b> – IRIM Seminar, Georgia Institute of Technology (US)      |
| 2025.02.10 | <b>Elevating Robotic Performance through Learned Autonomy</b> – ME Seminar, SNU (KR)              |
| 2024.10.15 | <b>Advancing the Autonomy of Legged Robots</b> – KAIST AI (KR)                                    |
| 2024.01.02 | <b>Learning locomotion and autonomous navigation for legged robots</b> – SUSTech (CN)             |
| 2023.12.29 | <b>Learning locomotion and autonomous navigation for legged robots</b> – Zhejiang University (CN) |
| 2023.08.02 | <b>Navigating real-world with legged robots</b> – Georgia Institute of Technology (US)            |
| 2022.11.10 | <b>Robust quadrupedal locomotion via RL</b> – Kraftron (KR)                                       |
| 2022.11.08 | <b>Toward the real-world applications of autonomous walking robots</b> – POSTECH (KR)             |
| 2022.11.03 | <b>Developing legged robots for outdoor use</b> – Agency for Defense Development (ADD) (KR)       |
| 2022.04.25 | <b>RL for quadrupedal locomotion</b> – DLR (DE)                                                   |
| 2022.01.12 | <b>Learning quadrupedal locomotion over challenging terrain</b> – RILAB, Korea University (KR)    |
| 2021.07.12 | <b>Learning quadrupedal locomotion over challenging terrain</b> – CRL, ETH Zurich (CH)            |

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| 2021.07.02 | <b>Learning quadrupedal locomotion over challenging terrain</b> – Telluride Neuromorphic Cognition Engineering Workshop (US) |
| 2021.05.21 | <b>Learning quadrupedal locomotion over challenging terrain</b> <a href="#">[video]</a> – NCCR Robotics Seminar (CH)         |
| 2021.02.26 | <b>Learning quadrupedal locomotion over challenging terrain</b> <a href="#">[video]</a> – PAIR, University of Toronto (CA)   |
| 2021.02.12 | 인공지능기술을 활용한 4족보행로봇 제어 – AI Winter School, Korean Society of Mechanical Engineers (KSME)(KR)                                  |
| 2020.11.10 | <b>Learning quadrupedal locomotion over challenging terrain</b> – CSIRO (AU)                                                 |
| 2020.08.05 | <b>Learning legged locomotion - walking through a forest</b> – Intel Labs (US)                                               |
| 2019.02.13 | 시뮬레이션에서의 심층강화학습을 이용한 보행로봇 제어 – Korea Institute of Industrial Technology (KR)                                                 |
| 2019.02.11 | 시뮬레이션에서의 심층강화학습을 이용한 보행로봇 제어 – Agency for Defense Development (ADD)(KR)                                                      |

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